MTH 200 Chapter 5: Integration

5.4 Applications of Integration

1. The height of a tree is currently 10 ft. Suppose the tree's height is increasing by $6\sqrt{t}\ in/month$ where t is measured in months from now. What will the trees height be in 4 months?

$$Height\ in\ 4\ months\ =\ current\ height\ +\ net\ change\ over\ 4\ months$$

$$=120in+\int_0^4 6\sqrt{t}dt=120+\left(4t^{3/2}\right)\Big|_0^4=120+\left[4(4)^{3/2}-4(0)^{3/2}\right]=120+32=152in$$

2. A bacteria population is 2000 at time $t=0$. At time t hours, the population is growing at a rate of $50t+100$ bacteria per hour. What is the population at time $t=2$ hours?

$$Pop.\ after\ 2\ hours=current\ pop.+net\ change\ over\ 2\ hours$$

$$=2000+\int_0^2(50t+100)dt=2000+\left[25t^2+100t\right]\Big|_0^2$$

$$=200+\left[25(2)^2+100(2)\right]-\left[25(0)^2+100(0)\right]=2000+300-0=2300\,bacteria$$

3. An oil storage tank ruptures and oil leaks from the tank at the rate of $r(t)=3\sqrt{2t}$ liters per minute where t is measured in minutes since the tank was ruptured. How much oil leaks out of the tank during the first 8 minutes?

$$Amount\ of\ oil\ leaked=\int_0^8 3\sqrt{2t}dt$$

Let $u=2t$ and thus $du=2dt \quad \rightarrow \quad \frac{1}{2}du=dt$

Our limits of integration become: $u(0)=2(0)=0 \quad \& \quad u(8)=2(8)=16$

$$Amount\ of\ oil\ leaked=\int_0^8 3\sqrt{2t}dt=\int_0^{16}\frac{3}{2}u^{1/2}du=\left(u^{3/2}\right)\Big|_0^{16}=(16)^{3/2}-(0)^{3/2}=64-0=64\,liters$$

4. Sal loves blueberries. At 9:00am, she started eating some at a rate of $\frac{480}{t}$ blueberries per minute, where t denotes the number of minutes since 9:00am. What is the total number of blueberries that Sal ate between 10:00am and 11:00am? Simplify your answer fully.

$$\textit{Number of blueberries eaten} = \int_{60}^{120} \frac{480}{t}\,dt = 480\ln|t|\Big|_{60}^{120} = 480\left[\ln(120) - \ln(60)\right]$$

$$= 480\ln\left(\frac{120}{60}\right) = 480\ln(2)\,\textit{blueberries}$$

5. The population of a town is currently 400, but it is expected to increase at a rate of $200e^{0.5t}$ people per year where t represents the number of years from now. What is the population of this town expected to be in 10 years?

Population in 10 years = Current population + net change over 10 years.

$$\textit{Population in } 10 \textit{ years} = 400 + \int_0^{10} 200e^{0.5t}\,dt$$

Let $u = 0.5t$ then $du = 0.5dt \quad \Rightarrow \quad 2du = dt$

Our bounds of integration become: $u(0) = 0.5(0) = 0 \quad \& \quad u(10) = 0.5(10) = 5$

Therefore,

$$\textit{Population in } 10 \textit{ years} = 400 + \int_0^{10} 200e^{0.5t}\,dt = 400 + 200\int_0^5 2e^u\,du = 400 + 400\int_0^5 e^u\,du$$

$$= 400 + 400\left[e^5 - e^0\right]$$

$$= 400 + 400e^5 - 400$$

$$= 400e^5$$

6. *BoxCo. Inc.* is a company that produces square boxes with side length x inches. The company is debating on increasing the side length of their boxes to $x+2$ inches, but is curious about the effect that this change would have on the volume of the boxes.

a) Represent the change in volume as a definite integral.

We know that if a side had length t, the volume of a *BoxCo.* box would be $V = t^3$ and therefore $\frac{dV}{dt} = 3t^2$.

Therefore we can represent the net change in the volume of the box with the integral $\int_x^{x+2} 3t^2\,dt$.

b) Calculate the change in volume (your final answer will be a formula in terms of x).

$$\textit{Change in Volume} = \int_x^{x+2} 3t^2\,dt = t^3\Big|_x^{x+2} = (x+2)^3 - x^3 \textit{ inches}^3$$

7. A particle moves with velocity $v(t) = 6\sin(3t)\, m/s$. Find the displacement and the distance travelled by the particle during $0 \le t \le \frac{\pi}{2}$.

Total displacement:

Let $u = 3t$. Then $du = 3dt \;\rightarrow\; \frac{1}{3}du = dt$. Our bounds become: $u(0) = 3(0) = 0$ & $u\left(\frac{\pi}{2}\right) = \frac{3\pi}{2}$

$$\textit{Total Displacement} = \int_0^{\pi/2} 6\sin(3t)\,dt = \int_0^{3\pi/2} 2\sin(u)\,du = -2\cos(u)\Big|_0^{3\pi/2}$$

$$= -2\cos\left(\frac{3\pi}{2}\right) + 2\cos(0) = 0 + 2 = 2\,meters$$

Total distance:

Note that:

$$6\sin(3t) = 0 \;\rightarrow\; 3t = 0 \;\rightarrow\; t = 0$$

$$OR \;\rightarrow\; 3t = \pi \;\rightarrow\; t = \frac{\pi}{3}$$

$$OR \;\rightarrow\; 3t = 2\pi \;\rightarrow\; t = \frac{2\pi}{3} \text{ (not in the given interval)}$$

So we have the intervals $\left(0, \frac{\pi}{3}\right)$ and $\left(\frac{\pi}{3}, \frac{\pi}{2}\right)$. We must determine if v is positive or negative on these.

$\left(0, \frac{\pi}{3}\right)$: Test $t = \frac{\pi}{4}$ $\qquad v\left(\frac{\pi}{4}\right) > 0$, so $v(t) > 0$ on the interval $\left(0, \frac{\pi}{3}\right)$.

$\left(\frac{\pi}{3}, \frac{\pi}{2}\right)$: Test $t = \frac{5\pi}{12}$ $\qquad v\left(\frac{5\pi}{12}\right) < 0$, so $v(t) < 0$ on the interval $\left(\frac{\pi}{3}, \frac{\pi}{2}\right)$.

Therefore, $\text{Total distance} = \int_0^{\pi/3} 6\sin(3t)\,dt - \int_{\pi/3}^{\pi/2} 6\sin(3t)\,dt$.

Let $u = 3t$. Then $du = 3dt \;\rightarrow\; \frac{1}{3}du = dt$. Our bounds become:

$u(0) = 3(0) = 0,\quad u\left(\frac{\pi}{3}\right) = 3\left(\frac{\pi}{3}\right) = \pi,\quad \&\quad u\left(\frac{\pi}{2}\right) = \frac{3\pi}{2}$

$$\text{Total distance} = \int_0^{\pi} 2\sin(u)\,du - \int_{\pi}^{3\pi/2} 2\sin(u)\,du = -2\cos(u)\Big|_0^{\pi} + 2\cos(u)\Big|_{\pi}^{3\pi/2}$$

$$= \left[-2\cos(\pi) + 2\cos(0)\right] + \left[2\cos\left(\frac{3\pi}{2}\right) - 2\cos(\pi)\right] = [2+2] + [0+2] = 6\,meters$$

8. A zombie outbreak has occurred in a major city at 9:00am and is spreading in a circular pattern from the initial point of infection at a rate $2^t\ mi^2 / hr$ where t denotes the number of hours since 9:00am.

   a) What is the total area (in square miles) the zombies have spread between 9:30am and 10:00am? Round your answer to the three decimals places and include the appropriate units.

$$Total\ Area = \int_{1/2}^{1} 2^t\,dt = \left.\frac{2^t}{\ln(2)}\right|_{1/2}^{1} = \frac{2^1}{\ln(2)} - \frac{\sqrt{2}}{\ln(2)} = \frac{2-\sqrt{2}}{\ln(2)} \approx 0.845\,mi^2$$

   b) You live 25 miles from the center of the outbreak. How long (starting from the time of the outbreak) will it take for the zombie infection to reach your house? Round your answer to the nearest minute.

We don't know how long it will take so suppose that x is the number of hours it takes for the disease to spread $625\pi mi^2$. Therefore, we can set up the following integral.

$$625\pi mi^2 = \frac{2^x - 1}{\ln(2)} \quad \rightarrow \quad 625\pi \ln(2) + 1 = 2^x$$

$$\rightarrow \quad \ln\left[625\pi \ln(2) + 1\right] = x\ln(2)$$

$$\rightarrow \quad x = \frac{\ln\left(625\pi \ln(2) + 1\right)}{\ln(2)} \approx 10.412\,hours \approx 10\,hours\,\&\,25\min.$$

Therefore, we estimate that the zombies will reach us after roughly 10 hours and 25 minutes.